

Exercises day 7

ClinVar

Disease	Description	Gene	Example Variant	Mutation Type	Functional effect
Hemochromatosis	Excess iron absorption (autosomal recessive)	HJV	NM_213653.3:c.959G>T	Missense	Likely LOF
Thalassemia	Reduced hemoglobin production	HBB	NC_000011.10:g.5225255_5225875del	Deletion	Likely pathogenic, mechanism unclear
Cystic Fibrosis	Thick mucus due to ion transport defect	CFTR	NM_000492.4(CFTR):c.4028dup (p.Cys1344fs)	Duplication (frameshift)	Pathogenic, causes nonsense-mediated decay and protein truncation
Tay-Sachs disease	GM2 ganglioside accumulation (neurodegeneration)	HEXA	NM_000520.6(HEXA):c.1559del (p.Gly520fs)	Deletion (frameshift)	Pathogenic, likely LOF

PDB

Superposition of WT hemoglobin (7VDE) and sickle cell hemoglobin (5E6E).

